

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5210
Subject Title	Space Vehicle Propulsion Systems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To equip students with an understanding of advanced thermodynamics and compressible flows, as well as their practical application in contemporary space vehicle propulsion systems, with a particular focus on rocketry.
Intended Learning Outcomes	Upon finishing this subject, students will be capable of: - Acquiring cutting-edge knowledge in aerospace propulsion systems including rockets, electric propulsion, and other future possible solutions. - Utilising their understanding, abilities, and practical experience to reasonably assess the performance of space propulsion systems. - Expanding their comprehension of space propulsion engineering to diverse engineering contexts and professional scenarios concerning space exploration. - Acknowledging the necessity for and demonstrating the capacity to participate in continual learning throughout their lives.
Subject Synopsis/ Indicative Syllabus	Introduction to propulsion: Air-breathing engines and rockets, applications of rocket propulsion Fundamentals of thermodynamics and compressible flows: Mass, momentum and energy conservation laws; First Law and second law of Thermodynamics; Ideal gas law; Entropy; Isentropic processes; Stagnation properties; Shock waves; Nozzle flows. Fundamentals of rocket propulsion: Thrust; Exhaust Velocity; Energy and Efficiencies; Multiple Propulsion Systems; Typical Performance Values; Variable Thrust. Liquid Propellant Rocket Engine: Types of Propellants; Tanks; Feed Systems; Gas Pressure Feed Systems; Tank Pressurisation; Turbopump Feed Systems and Engine Cycles; Rocket Engines for Maneuvering, Orbit Adjustments, or Attitude Control; Engine

	Families; Valves and Pipelines; Engine Support Structure; Liquid Propellant Combustion and Its Stability Solid Propellant Rocket Motor: Basic Relations and Propellant Burning Rate; Performance Issues; Propellant Grain and Grain Configuration; Propellant Grain Stress and Strain; Attitude Control and Side Manoeuvres with Solid Propellant Rocket Motors; Solid Propellants; Solid Propellant Combustion and Its Stability; Solid Rocket Motor Components and Design Hybrid Propellants Rocket Propulsion: Applications and Propellants; Design Example; Combustion Instability Electric Propulsion: Ideal Flight Performance; Electrothermal Thrusters; Nonthermal Electrical Thrusters; Optimum Flight Performance; Mission Applications; Electric Space-Power Supplies and Power-Conditioning Other Space Propulsion Systems: Nuclear Propulsion; Solar Sail Propulsion; Propellantless Propulsion (e.g., Space Elevator, Momentum Wheels)																																		
Teaching/Learning Methodology	The teaching and learning methods include lectures, mini-project and homework assignments. Technical/practical examples and problems will be raised and discussed in class. Project is designed to evaluate the space propulsion systems. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Mini-project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Homework assignments</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	✓	✓	✓	✓	Mini-project	✓	✓	✓	✓	Homework assignments	✓	✓	✓	✓										
Teaching/Learning Methodology	Outcomes																																		
	a	b	c	d																															
Lecture	✓	✓	✓	✓																															
Mini-project	✓	✓	✓	✓																															
Homework assignments	✓	✓	✓	✓																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Homework assignments</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Mini-project</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Final examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Homework assignments	25%	✓	✓	✓	✓	2. Mini-project	25%	✓	✓	✓	✓	3. Final examination	50%	✓	✓	✓		Total	100%				
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																															
		a	b	c	d																														
1. Homework assignments	25%	✓	✓	✓	✓																														
2. Mini-project	25%	✓	✓	✓	✓																														
3. Final examination	50%	✓	✓	✓																															
Total	100%																																		

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: $0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Final Examination}$ The continuous assessment, including homework assignments and project report serves to monitor students' progress, aid in self-assessment against the intended learning outcomes, and promote the integration of acquired knowledge from the lectures. Meanwhile, the final examination objectively evaluates students' knowledge acquisition, ability to comprehend and analyze problems critically and independently, and overall achievement of the subject learning outcomes.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	39 Hrs.
	Other student study effort:	
	▪ Self-study	67 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	1. Thermodynamics: An Engineering Approach, 8th Edition, 2014, Y. Cengel & M. Boles, McGraw-Hill Education 2. Fluid Mechanics: Fundamentals and Applications, 4th Edition, 2018, Y. Cengel & J. Cimbala, McGraw-Hill Education 3. Rocket Propulsion Elements, 9th Edition, 2016, G. Sutton & O. Biblarz, Wiley. 4. Elements of Propulsion: Gas Turbine and Rockets, 2nd Edition, 2006, J. Mattingl. & K. Boyer, AIAA.	

April 2024